

CI/CD

The Automatic Quality Checker

Building the Fast Track for Better Apps

Planrs Academy | Grade 6 Session 1

THE DAY THE APP CRASHED

A Developer's Nightmare

Imagine you are a developer for a world-famous game. You decide to add a cool new skin for the characters. It seems like a tiny change, so you hit "Upload" and go to bed.

The next morning, you wake up to millions of angry messages. The game is broken! A tiny spelling mistake in your code caused the whole app to crash for everyone.

The Problem: Humans make mistakes, especially when they are tired or in a hurry. If we have to check every single line of code ourselves, it takes too long. If we don't check it, we might break the app.

What if we had a **Robot Assistant** who could check our work automatically every time we changed something? This is where **CI/CD** comes in.

CI: CONTINUOUS INTEGRATION

The Automatic Gatekeeper

Continuous Integration (CI) is like a magic sensor at a gate. Every time a coder adds a new piece of code, the "CI Robot" immediately grabs it and runs it through a series of tests to see if it's correct.

The School Uniform Check:

Imagine your school has a magic gate sensor. As you walk through, it automatically checks:

- Is your tie straight?
- Are your shoes polished?
- Is your ID card on?

If everything is perfect, the gate opens with a "**Ding!**" (Successful Integration). If something is wrong, it "**Beeps**" and tells you exactly what to fix before you can enter.

FINDING MISTAKES EARLY

The Robot Pari-kshak

In CI, we call mistakes **Bugs**. The CI Robot is our best **Pari-kshak** (Tester). It checks the new code to make sure it doesn't fight with the old code.

What the CI Robot Checks:

- **Does the code run?** Or does it have a "Syntax Error" (like a spelling mistake)?
- **Does it break the game?** Will the characters still move correctly?
- **Is it tidy?** Is the code written clearly?

By catching these bugs early, we keep our game healthy and our users happy!

CD: CONTINUOUS DEPLOYMENT

The Instant Delivery

Once the CI Robot says "Pass!", what happens next? In the old days, a human would have to manually upload the new files. But with **Continuous Deployment (CD)**, the process is **Swachalit** (Automated).

The Automated Snack Factory:

Think of a factory making 'Haldiram's Bhujia.' It's a smooth, automated line:

1. One machine mixes the ingredients.
2. Another machine fries them.
3. Another machine packs them into packets.

As soon as a packet is sealed and passes the weight check, it automatically slides onto a delivery truck. No human has to carry each packet to the truck!

THE CI/CD PIPELINE

The Assembly Line for Code

The path that code takes from a developer's computer to a user's phone is called a **Pipeline**. It's a series of automated stages:

Stage 1: BUILD - The robot gathers all the code and "bakes" it into a real app.



Stage 2: TEST - The robot runs all the checks (The Magic Gate).



Stage 3: DEPLOY - The robot sends the "passed" app straight to the users.

This pipeline never sleeps! It works 24 hours a day to make sure updates are safe and fast.

WHY AUTOMATION IS OUR BEST FRIEND

Better, Faster, Stronger

Why do we let robots do the work? Here are the three main reasons:

1. Speed

Robots can check thousands of lines of code in seconds. Humans would take days!

2. Consistency

Robots don't get bored or distracted. They check the code exactly the same way every time.

3. Safety

Automatic testing acts as a safety net. It catches "Bugs" before they can reach the user and cause a crash.

VIRTUAL ACTIVITY

Build Your Pipeline

Imagine you are the Boss of the Snack Factory. You need to put these stages in the right order for your **CI/CD Pipeline**. Drag and drop these steps (mentally) into the right sequence:

[TEST THE QUALITY]

[DELIVER TO SHOP]

[MIX THE INGREDIENTS]

Correct Order:

1. Mix Ingredients (Build) -> 2. Test Quality (Test) -> 3. Deliver to Shop (Deploy)!

SUMMARY

The Core Lessons

- **CI (Continuous Integration):** The robot that automatically checks your code for mistakes (The Magic Gate).
- **CD (Continuous Deployment):** The robot that automatically sends your finished work to the user (The Snack Factory Truck).
- **Pipeline:** The automated path code takes from "Idea" to "App."
- **Success:** CI/CD means we can release new features fast without breaking things.

Success Criteria

Can you explain what a "Bug" is? Can you describe the three stages of a Pipeline? If yes, you have mastered the basics of CI/CD!

FINAL CHALLENGE

Show Your Knowledge

Before you go, think about this question:

"If the CI Robot finds a mistake in Stage 2 (TEST), should the code move to Stage 3 (DEPLOY)? Why or why not?"

Submit your thoughts to your teacher. Remember: In coding, automation is your superpower!